Code Explanation

Variable Declaration

```
int ldrPin = A0;
int ledPin = 13;
int ldrValue = 0;
```

- ldrPin stores the analog pin connected to the LDR sensor.
- ledPin stores the digital pin connected to the LED.
- ldrValue stores the light intensity value read from the LDR.

Setup Function

```
void setup()
{
 pinMode(ledPin, OUTPUT);
 Serial.begin(9600);
}
```

The setup() function runs only once when the Arduino starts.

- pinMode(ledPin, OUTPUT) configures the LED pin as an output.
- Serial.begin(9600) starts serial communication for monitoring sensor readings.

Reading Sensor Data

```
ldrValue = analogRead(ldrPin);
```

The Arduino reads the analog value from the LDR sensor. The value typically ranges from 0 to 1023 depending on the amount of light falling on the sensor.

Decision Making

```
if(ldrValue < 500)
{
 digitalWrite(ledPin, HIGH);
}
```

If the LDR value is less than 500, the environment is considered dark, and the LED is turned ON.

```
else
{
 digitalWrite(ledPin, LOW);
}
```

If the LDR value is greater than or equal to 500, sufficient light is present, and the LED is turned OFF.

Delay Function

```
delay(500);
```

The system waits for 500 milliseconds before taking the next reading. This prevents rapid switching and reduces unnecessary processing.